Source Verification and Claim Audit

Atlantic Canadian space infrastructure and data sovereignty

27 October 2026

The paper has undergone a fresh source audit, a clause-level review, and post-correction cross-review. The second audit revised 34 review units and added one methodological reference. The resulting paper has 34 references. Its central contribution remains a proposed diagnostic lens and research agenda; no interviews, local outcomes, ethics approval, or empirical validation are claimed.

What the audit establishes

The checks support the revised paper's representation of the evidence available to this review. They found and corrected an overgeneralization about measurement denominators, incomplete or imprecise dates and citation pinpoints, several categorical interpretations, and wording that implied roles or outcomes not established by the sources. No fabricated source was identified. This is an internal research audit, not external journal peer review, a legal opinion, a compliance inspection, community validation, or an ethics determination.

How the three stages were performed

Source identity and primary evidence

Each cited work was checked against its primary publication, official record, or original organizational statement. Relevant passages, dates, authorship, identifiers, jurisdiction, draft status, and source type were examined. Direct-access failures were recorded. A primary search index or abstract was not treated as a complete fresh full-text read.

Claim and interpretation review

Six separate review assignments covered all substantive body paragraphs and all table rows. Statements were classified as documented facts, interpretations, proposals, normative premises, methods, questions, or evidence limits. Compound units were read clause by clause. Reviewers checked whether a source entailed the claim and whether the interpretation exceeded its population, time, geography, purpose, or study design.

Corrections and cross-review

All 34 amended units were cross-reviewed by a reviewer who did not originate the relevant recommendations, followed by a final reconciliation. That pass clarified two statutory references and restored an antecedent in a proposed-constraints sentence. Bibliography matching, coverage, source access records, document identity, and rendered pages were checked. These stages do not mean every underlying dataset or every reference inside each source was independently reanalysed.

Coverage and classifications

The detailed register contains 391 substantive review units spanning all body prose and table rows. Units are sentences, compound statements, or table rows, not 391 independent empirical claims. A sentence-indexing fragment containing only a citation continuation was merged with its preceding unit; no

substantive content was dropped. The labels and coverage were manually reconciled after the initial indexing step. Titles, dates, captions, bibliography, and document structure were checked separately.

Statement type	Review units
Methodological statement	37
Proposal	127
Source interpretation	83
Review limitation	35
Question	14
Normative premise	59
Documented fact	36
Total	391

A supported proposal means it is correctly identified and reasoned as a proposal. It does not mean its effectiveness has been measured. Likewise, a bounded interpretation can be defensible without being the only possible interpretation. The spreadsheet register preserves these distinctions instead of assigning one universal truth score.

Corrections and qualifications

The table identifies every amended unit. Stable review IDs locate entries in the accompanying filterable register. Locations refer to paper sections; the register also gives a short text anchor and the relevant source IDs. These brief labels avoid duplicating the paper's source paraphrases across deliverables.

Unit	Section	Revision purpose
C035	1	Separate requirements from intentions
C044	2	Use undated continuing register
C046	2	Retain dated approval
C054	2	Specify pad and associated services
C065	2	Preserve prospective provider wording
C067	2	Separate hosting from data authority
C099	3	Do not presume claim legitimacy
C101	3	Condition duties on relevant activity
C129	3	Respect scope of binding authority
C147	4	Specify distinct statutory duty holders
C151	4	Specify personal and employer scope
C169	4	Identify proposed recommendation

Unit	Section	Revision purpose
C174	4	Eligibility does not establish control
C177	4	Bound gap to selected cases
C179	4	Uncertainty motivates proposed inquiry
C186	4	Do not presume future mandates
C212	5	Preserve perspective limitation
C214	5	Attribute organizational description
C215	5	Qualify present operational status
C221	5	Distinguish counts from rates
C229	5	Retain constraints as hypotheses
C237	5	Avoid impossible local extrapolation
C250	6	State transfer and expertise limits
C260	6	Do not presume representation
C270	6	Evaluate fidelity and lost context
C274	6	Use displayed update date
C297	7	Add authority diversity pinpoints
C300	7	Narrow exemption statement
C301	7	Clarify present draft review status
C303	7	Cite exploratory phase rule
C308	7	Frame plural evidence as recommendation
C319	7	Justify sample adequacy
C340	7	Bound interview validation claims
C384	9	Keep pathway as research proposal

Source verification register

The entries report this fresh pass, not an assumption that the earlier access status remained unchanged. All 34 works were rechecked at the level stated. TCPS 2 Chapters 9 and 10 are parts of one cited policy; Paolo's correction was checked alongside the article. Most relevant material was available as complete official or publisher text. The access limitations below are retained wherever a complete fresh read was not possible. Access date: 27 October 2026.

01 Canada, 2000

Material checked: Fresh official full HTML, including definitions, application and relevant Schedule 1 clauses; not just prior indexed text.

Locator: Sections 2, 4(1)–(2),4.01, 5–9; Schedule 1 clauses 4.1–4.3, 4.7, 4.9–4.10

Claim group: Personal-information definition, covered commercial and specified employment activities, and selected processing principles.

Boundary: No operator classified as a federal undertaking; no project compliance finding.

https://www.laws.justice.gc.ca/eng/acts/P-8.6/FullText.html

02 Canada, 2005

Material checked: Fresh official full HTML and targeted relevant provisions.

Locator: Sections 2, 4–8, 14–15

Claim group: Scope, licensing, system and data control, service interruption and priority-access powers.

Boundary: Country-government access in s8(4)(c) distinguished from local community access; lawful limits retained.

https://lois-laws.justice.gc.ca/eng/acts/R-5.4/FullText.html

03 Canada, 2021

Material checked: Fresh official full HTML and targeted operative provisions.

Locator: Sections 4–6, including 6(1); Schedule

Claim group: Federal consistency measures, consultation and cooperation, action plan and implementation framework.

Boundary: Consistency and action-plan duties are actor-specific; no complete project protocol inferred.

https://laws.justice.gc.ca/eng/acts/U-2.2/FullText.html

04 Canadian Institutes of Health Research et al., 2022

Material checked: Fresh complete official Chapters 9 and 10 HTML; relevant articles and application text directly inspected.

Locator: Chapter 9, arts. 9.1–9.3, 9.5–9.7, 9.11, 9.13–9.17; Chapter 10, art. 10.1 and Application.

Claim group: Community engagement, authority and diversity, research agreements, individual consent, benefits, capacity, confidentiality and interpretation.

Boundary: Ethics policy has contextual rules and exceptions; it is not a space-sector statute, universal leadership veto, or evidence that this paper received approval.

https://ethique.gc.ca/eng/tcps2-eptc2_2022_chapter9-chapitre9.html

https://ethics.gc.ca/eng/tcps2-eptc2_2022_chapter10-chapitre10.html

05 Canadian Space Agency [CSA], 2022; CSA, 2022

Material checked: Fresh official strategy overview and vision webpages.

Locator: About this strategy, What we heard; vision, Whole-of-society engagement

Claim group: Policy commitment to Indigenous-community collaboration and open, accessible satellite Earth observation data.

Boundary: January 20 2022 is displayed modification date; policy commitment is not a demonstrated local outcome.

https://www.asc-csa.gc.ca/eng/publications/canada-strategy-for-satellite-earth-observation/05-about-this-strategy-a-bold-leap-into-the-future.asp

06 CSA, 2026

Material checked: Fresh complete official call HTML; publication date and relevant eligibility verified directly.

Locator: Publication date August 31 2026; sections2, 3.1, 3.2

Claim group: Applications for nontraditional users including Indigenous communities; eligible recipients limited to Canadian universities and post-secondary institutions.

Boundary: No guarantee governance-only research qualifies; institutional eligibility does not establish unilateral governance of a partnership.

https://www.asc-csa.gc.ca/eng/funding-programs/funding-opportunities/ao/2026-research-opportunities-in-satellite-earth-observation.asp

07 Carroll et al., 2020

Material checked: Fresh full publisher HTML succeeded after an initial timeout; authors, article type, date and relevant text inspected.

Locator: Abstract; Background; Emergence of Indigenous Data Sovereignty; The CARE Principles and Their Supporting Concepts; publisher metadata

Claim group: Collective Benefit, Authority to Control, Responsibility and Ethics; relationship to FAIR and people-centred governance.

Boundary: Practice/principles paper; does not establish local uptake or quantified benefits.

https://doi.org/10.5334/dsj-2020-043

08 Department of National Defence, 2026

Material checked: Official complete HTML reopened; date, agreement paragraph and Quick facts checked.

Locator: Date line; Canadian-owned spaceport; Launch the North; Quick facts.

Claim group: Announced ten-year $200 million agreement to lease a pad, end-2026 capability requirement, separate 2028 light-lift target and Canadian expenditure requirement.

Boundary: Targets and announced contract terms; no verified delivered orbital service or local allocation.

https://www.canada.ca/en/department-national-defence/news/2026/03/minister-mcguinty-announces-strategic-investments-in-sovereign-space-launch.html

09 European Union, 2016

Material checked: Direct EUR-Lex HTML and PDF bot-blocked. Official indexed text for Articles 2–6 was freshly retrieved; UK official adopted-text index corroborated Articles 2–5.

Locator: Articles 2–6

Claim group: Personal-data focus, material and territorial reach, processing principles and lawful bases.

Boundary: No claim of full consolidated-text manual review. Only verified provisions used; no current project-specific GDPR opinion.

https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679

10 First Nations Information Governance Centre [FNIGC], n.d.; FNIGC, n.d.

Material checked: Fresh full official HTML opened; definitions and FAQ inspected.

Locator: Understanding OCAP; First Nations-specific FAQ; Trademark; How do I cite OCAP; Learning about OCAP

Claim group: First Nations-specific principles, their meanings and interpretation through Nations' own worldviews and protocols.

Boundary: Normative principles and scope; not endorsement of a particular system.

https://fnigc.ca/ocap-training/

11 Groth & Moreau, 2013

Material checked: Official W3C overview freshly reopened; metadata, Abstract, status and roadmap checked. PROV-DM Recommendation cross-checked for entities, activities and agents.

Locator: Header; Abstract; Status; Introduction; PROV-DM Abstract at https://www.w3.org/TR/prov-dm/.

Claim group: Interoperable representation of provenance involving entities, activities and agents.

Boundary: Overview is explicitly a non-normative Working Group Note, not a Recommendation. Its family includes Recommendations. Recorded provenance is not proof of truth or consent.

https://www.w3.org/TR/prov-overview/

12 Hummel et al., 2021

Material checked: Fresh full publisher HTML opened; methods, discussion, Table 5 and publication metadata inspected.

Locator: Methods; Discussion and Table 5; Information and rights metadata

Claim group: Plural meanings of sovereignty across agents, contexts, values, rights and practical abilities; need for explicit specification.

Boundary: Maps meanings; does not evaluate a preferred architecture. January 22, 2021, volume 8 issue 1 confirmed.

https://doi.org/10.1177/2053951720982012

13 Iannella & Villata, 2018

Material checked: Official W3C Recommendation freshly reopened; status, editors, date, Abstract, Aims, rule classes and Privacy Considerations checked.

Locator: Header; Abstract; sections 1.1, 2.5–2.6 and 4.

Claim group: Policy expression of permissions, prohibitions, duties and constraints.

Boundary: Policy expression and processing conformance do not establish legitimate authority or legal compliance. No guarantee against off-system use.

https://www.w3.org/TR/odrl-model/

14 Innovation, Science and Economic Development Canada [ISED], 2026; ISED, 2026

Material checked: Fresh official consultation HTML, including Next steps and modification date.

Locator: Paragraphs 1–2, 20–27, 80–89; Q26

Claim group: Modernization consultation scope including ground stations and data, possible Space Act and coordination, and December 21, 2026 comment deadline.

Boundary: Consultation options are not enacted powers or adopted community-service arrangements.

https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en/learn-more/key-documents/consultations/consultation-legislative-and-regulatory-modernization-space-activities

15 Jennings et al., 2025

Material checked: Fresh direct publisher HTML/PDF hit cookie redirects; primary publisher indexed abstract and relevant full-text sections retrieved, with PubMed indexed bibliographic record corroboration. Do not call this a fresh direct full-article read.

Locator: Publisher abstract, article-type/date metadata and NEON discussion; PubMed PMID 39794335

Claim group: Relevance of Indigenous governance to environmental information and the design of permissions, protocols, metadata and infrastructure.

Boundary: Perspective, published 10 January 2025, Nature Communications 16 article 572; authors and DOI confirmed. Manuscript uses only broad scope/type claims supported by retrieved primary text.

https://doi.org/10.1038/s41467-024-53480-2

16 Leaf Space, 2026

Material checked: Provider announcement HTML reopened; date and both substantive paragraphs checked.

Locator: September 8, 2026 date; completion statement; prospective support statement.

Claim group: Provider's announced completion of a Canadian ground station and described S- and X-band support.

Boundary: Provider announcement; support stated prospectively, no independent service or governance validation.

https://leaf.space/leaf-space-extends-its-global-network-with-first-ground-station-in-canadas-spaceport-nova-scotia-2/

17 Local Contexts, 2026

Material checked: Organization FAQ freshly reopened; displayed Last Updated September 10, 2026 and relevant questions directly checked.

Locator: Last Updated; Labels versus Notices; copyright/Creative Commons; ownership/customization of Labels.

Claim group: Community-controlled customization and use of Labels; extra-legal mechanisms complementing existing licensing and law.

Boundary: Community-selected tools; FAQ calls them extra-legal mechanisms. Description does not supply legal or technical enforcement. Hub is not a data repository.

https://localcontexts.org/support/frequently-asked-questions/

18 Malterud et al., 2016

Material checked: Primary publisher abstract and publication record, PubMed abstract/metadata, and author university record checked; full article text not inspected.

Locator: Abstract and issue metadata.

Claim group: Sample adequacy depends on study design and information power, not a universal count.

Boundary: Conceptual guidance; full article not inspected and no twenty-interview threshold established.

https://doi.org/10.1177/1049732315617444

19 Mamun and Natcher, 2023

Material checked: Fresh publisher full HTML examined on 27 October 2026, including Abstract, Materials and methods, Challenges, Conclusion and citation metadata.

Locator: Abstract; Materials and methods final paragraph; Challenges; Conclusion; Cite this article.

Claim group: Potential of community monitoring alongside funding, training, stewardship and data-use constraints.

Boundary: Narrative review of heterogeneous records; its authors explicitly distinguish researchers' outcome accounts from community perspectives. Not a local spaceport evaluation.

https://doi.org/10.1007/s10661-022-10841-y

20 Maritime Launch Services, 2026

Material checked: Company notice HTML reopened; date, participation list and advertised committee functions checked.

Locator: September 3, 2026 date; local voices; advisory functions.

Claim group: Advertised participation by fishers, residents, businesses and local organizations, and described advisory communication role.

Boundary: Recruitment and advertised role, not assessed representative legitimacy or influence.

https://www.maritimelaunch.com/news/join-community-liaison-committee-2026

21 Mi'kmaw Conservation Group, n.d.

Material checked: Fresh organization webpage fully opened on 27 October 2026. Introductory role/board statement and project lists checked; page has no modification date and includes 2020–2021/2020 records.

Locator: Mi'kmaw Conservation Group At Work In Communities; Oceans Projects; Watersheds Projects; Internal Progress Reports.

Claim group: Described community technical and monitoring capacity guided by fisheries managers, rights holders and Elders.

Boundary: Self-description only; current staffing, specific project status and willingness to partner were not independently established.

https://mikmawconservation.ca/programs-and-services/current-projects/

22 Mi'kmaw Ethics Watch, n.d.

Material checked: Fresh full current application-page HTML and FAQs opened on 27 October 2026.

Locator: FAQs on university REB review, exemptions and research without human participants, lines 135–166.

Claim group: Need to confirm applicable Mi'kmaw review and exemption processes alongside institutional ethics requirements.

Boundary: Potential exemption categories are not automatic proof of this draft's status. Wording should distinguish an exemption inquiry when uncertain from a universal assertion that every exemption needs formal assessment.

https://mawkim.org/mikmaw-ethics-watch/apply-for-research-approval/

23 Nova Scotia, 2026

Material checked: Fresh official October 1 2026 news-release HTML.

Locator: Quick Facts, second bullet; current/replacement Act links

Claim group: Replacement Freedom of Information and Protection of Privacy Act scheduled to take effect April 1, 2027.

Boundary: Used for announced commencement only, not privacy-performance statistics.

https://news.novascotia.ca/en/2026/10/01/nova-scotia-among-canadas-leaders-foipop-response-rates

24 Nova Scotia Department of Justice, n.d.

Material checked: Fresh official FAQ HTML; current government overview also checked. Direct statute fetch still timed out.

Locator: What is PIIDPA; public body and service-provider definitions; outside-Canada requirements and exceptions

Claim group: PIIDPA public-body, municipal and service-provider scope and qualified Canadian storage/access baseline.

Boundary: Guidance, not a substitute for case-specific statutory interpretation. Historical FAQ commencement statements not relied upon.

https://novascotia.ca/just/iap/piidpaquest.asp

25 Nova Scotia Environment, n.d.

Material checked: Official full register reopened; indexed register independently searched.

Locator: Highlights; Description; Project Update; Minister’s Decision.

Claim group: Approval with conditions and June 4, 2019 decision; retained historical project and schedule assumptions.

Boundary: Historical planning information; no current manifest or compliance finding.

https://novascotia.ca/nse/ea/canso-spaceport-facility/

26 Nova Scotia Environment, 2019

Material checked: Official 14-page scanned PDF reopened; pages 1, 2, 5 and 11 visually inspected; relevant text cross-checked.

Locator: Approval date on page 1; condition 3.3 page 2; 4.8–4.9 page 5; 9.2, 9.4, 10.1 page 11.

Claim group: Compliance reporting, environmental plans and monitoring, complaint route, community liaison committee, and Mi'kmaq communication plan at the Department's request.

Boundary: Conditions alone do not prove compliance, implementation or consent.

https://novascotia.ca/nse/ea/canso-spaceport-facility/Conditions.pdf

27 OECD, 2026

Material checked: Official report landing page and Chapter 3 freshly reopened; date, DOI, title, corporate author and enforcement passages directly checked.

Locator: Landing metadata 20 October 2026; Chapter 3 sections 3.3.3 and 3.4; notes 14–16.

Claim group: Coordination and viability constraints; limitations of post-transfer usage controls; complementary contracts, provenance and audit.

Boundary: Policy analysis using initiative registry, cases, workshop and theory. No Atlantic field test. Report itself flags reporting bias and hypotheses in some discussions; cited limited enforcement point is explicit.

https://doi.org/10.1787/0066f11b-en

28 Paolo et al., 2024

Material checked: Fresh publisher and PMC full-page/PDF opens were blocked by redirects or CAPTCHA. Publisher-indexed original text, indexed PMC Methods, and official bibliographic records were examined; this fresh pass did not inspect a complete primary article.

Locator: Abstract; Main AIS requirements/public visibility; indexed Methods; publisher citation metadata.

Claim group: Usefulness of combined imagery and vessel information and limits of public AIS observation and interpretation.

Boundary: Supports the narrow integration/AIS statements. Global industrial-vessel results do not validate small-vessel performance near Canso.

https://doi.org/10.1038/s41586-023-06825-8

29 Paprica et al., 2023

Material checked: Publisher HTML freshly reopened. Abstract, author list, publication sidebar, Governance, Approach and aims, and Limitations directly examined.

Locator: Abstract; Governance; Limitations and future work; publication sidebar.

Claim group: Operational requirements tested and refined with 23 organizations; authority, accountability, management, engagement and links between governing bodies and data decisions.

Boundary: Organizational refinement rather than causal evaluation. Mostly Canada/USA public or not-for-profit contexts; authors acknowledge lack of detailed Indigenous operational expertise and need for commercial-sector work. DOI/year/title/26-author order confirmed.
https://doi.org/10.23889/ijpds.v8i4.2142

30 Strum Consulting, 2018

Material checked: Official-hosted 159-page proponent PDF reopened; cover, section 8 and physical/printed page mapping checked.

Locator: Cover and submission letter; section 8 printed pages 107–108, physical pages 132–133.

Claim group: Historical proponent account of Membertou Geomatics Solutions ecological knowledge study and knowledge-holder interviews during November 2017–January 2018.

Boundary: Proponent account; not an independent consent or reuse-authority determination.
https://novascotia.ca/nse/ea/canso-spaceport-facility/registration_document.pdf

31 Transport Canada, n.d.

Material checked: Official draft HTML reopened; opening disclaimer and version table checked.

Locator: Opening draft notice; Table of Versions ending April 30, 2025, version 0.4.

Claim group: Linked application guidance remained explicitly marked draft at the review date.

Boundary: No full-document compliance review; claim limited to draft status.
https://tc.canada.ca/en/aviation/commercial-space-launches-canada/application-requirements-obtaining-authorization-launch-uncrewed-orbital-suborbital-rockets-canada

32 Transport Canada, 2026a

Material checked: Official complete overview HTML reopened.

Locator: Site selection; launch permission; displayed April 23, 2026 modification date.

Claim group: Transport Canada does not license spaceports; launch applicants require ministerial permission and must establish site suitability.

Boundary: Commercial-launch overview, not project-specific authorization evidence.
https://tc.canada.ca/en/aviation/commercial-space-launches-canada

33 Transport Canada, 2026b

Material checked: Official complete consultation HTML reopened.

Locator: Closed status; consultation dates; draft requirements; public-notification question; modification date.

Claim group: Feedback on draft application requirements, public launch notification and future regulation-making.

Boundary: Closed consultation and proposed requirements, not adoption or enacted rules.
https://tc.canada.ca/en/corporate-services/consultations/consultation-commercial-space-launch-canada

34 United Nations, 2007

Material checked: Fresh official statutory Schedule HTML opened and articles inspected; Act consolidation current to 30 September 2026.

Locator: Schedule articles 3, 18–19, 31–32 and 45; Declaration adoption identified in Act section 2

Claim group: Participation through Indigenous representative institutions, cultural heritage and knowledge, and state consultation/cooperation to obtain consent in the specified circumstances.

Boundary: State-directed provisions and collective rights; application to project data remains contextual.
https://laws.justice.gc.ca/eng/acts/U-2.2/FullText.html

Limits that remain

Fresh direct full-text retrieval remained restricted for the GDPR, Jennings et al., and Paolo et al.; official or publisher-indexed relevant passages and bibliographic records were checked and the corresponding uses are narrow. The PIIDPA discussion cites the official guidance actually read; it does not imply direct

reinspection of the entire statute. Malterud et al. is used only for the sample-planning principle supported by its primary abstract and metadata. These access limits should not be described as exhaustive full-text verification.

The paper's desk-review and subsequent empirical research requirements should be confirmed with MEW and the relevant institution before publication or the relevant research activities. No approval or exemption determination was obtained in this audit. The public nature of a document does not itself settle that status. Community preferences, private agreements, operational performance, net benefit, and framework validity remain empirical questions.

The search is targeted rather than exhaustive. It cannot establish that unlocated research or agreements do not exist. Legal or project-specific implementation requires the relevant current instruments and appropriate advice. A source check cannot turn a design proposal into a demonstrated outcome or guarantee that the paper contains no remaining error.

How to use the detailed register

Open Atlantic_Space_Claim_Register.xlsx. Filter the Claims sheet by category, action, section, or source ID. Each row gives a short text anchor for finding the relevant passage in the revised paper, the audit disposition, and an evidence locator. The Sources sheet stores the checked reference and access boundary once. Original review IDs are retained, so there is a deliberate gap at C183, the merged citation continuation. Summary counts describe reviewed units, not verified empirical findings.

Final document identity

Paper file: Atlantic_Space_Data_Sovereignty_Paper.docx

SHA256: cc52a48abadb5402356e3a14f53293208c6f976a56e68bd277cd78b406d0b649